STATS 765 - Project Milestone 04

Youth Survey Response

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1 Goal

In this project, we will develop a classifier to predict which respondents in the NLSY97 dataset are likely to be unreliable based on their response patterns. This involves exploring relevant variables, developing labeling rules, and using machine learning algorithms to build a reliable classifier for improving the accuracy and reliability of the NLSY97 dataset.

2 Data Source

The National Longitudinal Survey of Youth 1997 (NLSY97) dataset can be obtained from the U.S. Bureau of Labor Statistics (BLS). The dataset, which follows a nationally representative sample of young men and women born in the years 1980-1984 and living in the United States when the survey began in 1997, was downloaded as a `.dat` file. For the purpose of this project, we focused on variables related to respondents' demographic information and response patterns.

3 Data Processing

In this section, we will discuss the data processing steps taken to address the identified data issues, such as handling missing values, encoding categorical variables, and normalizing numerical variables. We will also detail the selected features from the NLSY97 dataset and the rationale behind their choice. Since we are dealing with survey data, missing values will be left as is to maintain the integrity of the responses.

3.1 Selected Features

From the NLSY97 dataset, we chose the following features:

1. Employment status (E0012001, E0012101): These variables can be used to understand the reliability of respondents by assessing their engagement in the workforce.
2. Attitude-related variables (R0357600, R0357700, R0357800): These variables can help identify respondents' outlook on life, which may influence their reliability in answering survey questions.
3. Criminal behavior variables (R0514800, S0932600, S2995600): These features can provide insights into the respondents' propensity for dishonesty, which may affect their reliability.
4. Mental health variables (R0690500, R0690800, R0691000, R0691200, S0921200): These variables can help assess the respondents' mental health, which may influence their reliability in answering survey questions.
5. Demographic variables (R0536300, R1235800, R1482600): These features can be used to identify potential patterns among different demographic groups in terms of reliability.
6. Education-related variables (R1302400, R1302500, R1302600, R1302700, R9871900, Z9083900): These variables can provide insights into the respondents' educational background, which may correlate with their reliability.

These factors can potentially influence the respondents' reliability when answering survey questions.

3.2 Load the dataset and handle missing values

Before processing the categorical and numerical variables, we first address the missing values in the dataset. In the NLSY97 data, specific codes represent different types of missing values. We will convert these codes to NA to represent missing data in a standard format.

```
data <- read.table("NLSY97.dat", sep = " ")
data <- data[, -3]  # remove case identification code
data[data == -1] = NA  # Refused
data[data == -2] = NA  # Dont know
data[data == -3] = NA  # Invalid missing
data[data == -4] = NA  # Valid missing
data[data == -5] = NA  # Non-interview
```

The dataset contains 8984 observations and 50 features.

3.3 Encoding categorical variables

Categorical variables in the dataset, such as gender, race, and ethnicity, were encoded as factors to facilitate their use in machine learning algorithm.

```
# Encoding categorical variables
d = vallabels(data)

# Rename variables using Qnames instead of Reference
# Numbers
d = qnames(d)
```

3.4 Data Labeling

We have created a new column `class` in our dataset to store the labels for whether a survey respondent is reliable (0) or unreliable (1). We used a set of custom functions (`consistency` and `increase`) to compare the features listed below and flagged the respondents as unreliable if inconsistencies were found:

- `MONTH AND YEAR FIRST HAD SEX (2000 and 2001)`: The month and year when the respondent first had sex.
- `HAS R BEEN ARRESTED SDLI (2002 and 2003)`: Whether the respondent has been arrested.
- `R EVER HAVE SEXUAL INTERCOURSE (2002 and 2003)`: Whether the respondent ever had sexual intercourse.
- `R HAD SEX > ONCE? (2002 and 2003)`: Whether the respondent had sex more than once.
- `SEX PARTNERS R HAS EVER HAD? (2002 and 2003)`: The number of sexual partners the respondent ever had.
- `HAS R EVER BEEN PREGNANT? (2002 and 2003)`: Whether the respondent ever been pregnant.
- `EVER USE COMPUTER TO GET TO WEBSITE? (2003 and 2004)`: Whether the respondent ever used a computer to access a website.
- `HOW MANY TIMES R DRIVE WHEN DRINKING ALCOHOL (2007 and 2008)`: How many times the respondent drove a vehicle after drinking alcohol.

```
# Mark unreliable samples
for (i in 1:nrow(d)) {
    if (!consistency(d$`YSAQ-312M~M_2000`[i], d$`YSAQ-312M~M_2001`[i]) |
        !consistency(d$`YSAQ-312M~Y_2000`[i], d$`YSAQ-312M~Y_2001`[i]) |
```

```
        !increase(d$`YSAQ-441_2002`[i], d$`YSAQ-441_2003`[i]) |
        !increase(d$`YSAQ2-299_2002`[i], d$`YSAQ2-299_2003`[i]) |
        !increase(d$`YSAQ2-303_2002`[i], d$`YSAQ2-303_2003`[i]) |
        !increase(d$`YSAQ2-304_2002`[i], d$`YSAQ2-304_2003`[i]) |
        !increase(d$`YSAQ2-314_2002`[i], d$`YSAQ2-314_2003`[i]) |
        !increase(d$`YSAQ-INTRO-2_2003`[i], d$`YSAQ-INTRO-2_2004`[i]) |
        !increase(d$`YHEA-810_2007`[i], d$`YHEA-810_2008`[i])) {
        d$class[i] <- 1  # Unreliable
    } else {
        d$class[i] <- 0
    }  # Reliable
}
```

The result is that there are 775 unreliable samples in the dataset.

3.5 Identify relevant features

We first remove the features that we have used for manual labeling from the dataset `d` and store the resulting dataset in `d2`.

To identify the features that are most relevant to determining whether a respondent is reliable or unreliable, we perform Recursive Feature Elimination (RFE) using Random Forest as the feature selection algorithm to identify the top 20 most relevant features for predicting. We performed 5-fold cross-validation to estimate the performance of the process. The function cannot handle missing values by default. In this case, we use `na.roughfix` function to handle the missing values temporarily.

```
set.seed(seed)
d2 <- na.roughfix(d2)
control <- rfeControl(functions = rfFuncs, method = "cv", number = 5,
    verbose = FALSE)
num_features_to_retain <- 20
rfe_results <- rfe(d2[, -which(names(d2) %in% "class")], d2$class,
    sizes = num_features_to_retain, rfeControl = control)
```

The top 20 selected features are:

```
##  [1] "CVC_WKSWK_ADULT2_ET_XRND"     "CVC_WKSWK_ADULT2_ALL_XRND"
##  [3] "CV_YTH_REL_HH_CURRENT_2003"   "CVC_HIGHEST_DEGREE_EVER_XRND"
##  [5] "CV_INCOME_GROSS_YR_2003"      "CV_HGC_RES_DAD_1997"
##  [7] "CV_HGC_RES_MOM_1997"          "KEY_RACE_ETHNICITY_1997"
##  [9] "YHEA-100_2003"                "CV_HGC_BIO_DAD_1997"
## [11] "CV_HGC_BIO_MOM_1997"          "CVC_WKSWK_TEEN2_XRND"
## [13] "YSAQ-282G_2002"               "EMP_STATUS_2001.01_XRND"
## [15] "TRANS_CRD_GPA_OVERALL_HSTR"   "CV_YTH_REL_HH_CURRENT_1997"
## [17] "KEY_BDATE_Y_1997"             "EMP_STATUS_2000.01_XRND"
## [19] "CV_ESR_2000"                  "YHEA-100_2000"
```

Then we create a new dataframe called `d3` that contains only the top 20 selected features and the `class` column for further analysis, model building, or validation.

```
d3 <- d[, c(selected_features, "class")]
```

4 Analytical Plan

4.1 The Classifier Model

We use the XGBoost algorithm to build a classifier model that can predict which respondents are likely to be unreliable based on their response patterns and other relevant features. It can handle missing values and automatically learns the best features for the classification task.

We split the data into training and testing datasets and convert the training and testing datasets into `xgb.DMatrix` format. We then perform cross-validation to determine the optimal parameters for the XGBoost model and show the result:

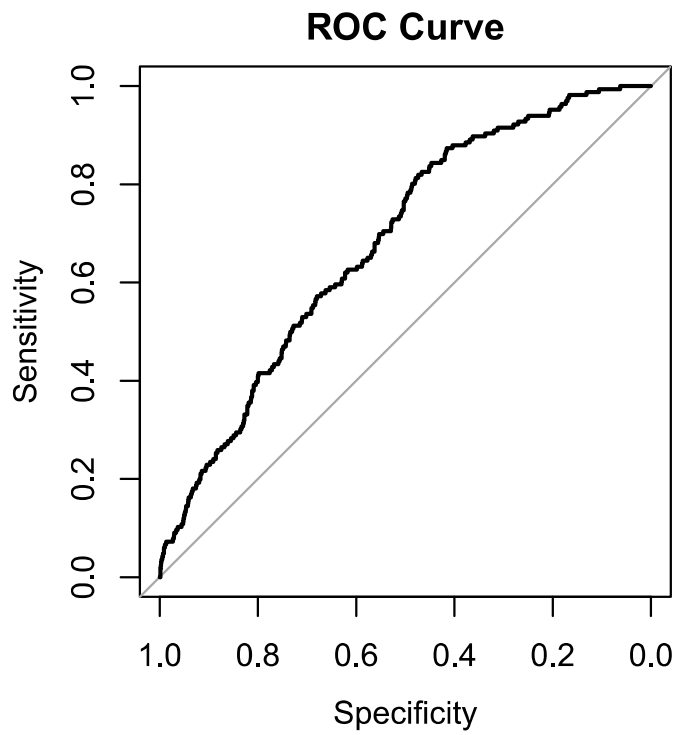
```
set.seed(seed)
# Define the parameter grid
params_grid <- expand.grid(max_depth = c(2, 4, 6), eta = c(0.01,
    0.1, 0.3), gamma = c(0, 0.1, 0.2), colsample_bytree = c(0.6,
    0.8, 1), min_child_weight = c(1, 3, 5), subsample = c(0.6,
    0.8, 1))

# Define the number of folds for cross-validation
n_folds <- 5

# Run cross-validation with hyperparameter tuning
cv_results <- list()
for (i in 1:nrow(params_grid)) {
    params <- as.list(params_grid[i, ])
    params$objective <- "binary:logistic"
    params$eval_metric <- "logloss"
    params$nthread <- 1
    result <- xgb.cv(params = params, data = train_data_matrix,
        nrounds = 20, nfold = n_folds, early_stopping_rounds = 10,
        verbose = 0, stratified = TRUE)
    cv_results[[i]] <- list(params = params, result = result)
}

# Find the best model based on the evaluation metric
# (logloss)
best_model <- cv_results[[which.min(sapply(cv_results, function(x) min(x$result$evaluation_log
```

After obtaining the optimal parameters, we train the XGBoost model:

```
best_params <- best_model$params
final_model <- xgboost(data = train_data_matrix, params = best_params,
    nrounds = 100)
test_predictions <- predict(final_model, test_data_matrix)
```

We evaluate the model on the test dataset:

```
## Area under the curve: 0.6806
```



The test accuracy is 0.9075724. Then we plot the ROC curve and calculate AUC scores, which is 0.6806046.

Based on the results above and our current understanding of the dataset, we propose the following analytical method:

1. Classifier Model: We will continue to use the XGBoost algorithm for building the classifier model, as it has shown promising results during our data exploration.
2. Handling Imbalanced Data: Since the dataset is imbalanced, with most respondents being reliable, we need to address this issue to avoid biased predictions:
   - Resampling: We can either oversample the minority class (unreliable respondents) or undersample the majority class (reliable respondents) to create a more balanced dataset. In our case, most respondents in the dataset are classified as reliable, leading to an imbalance between reliable and unreliable classes. To address this issue, we can apply resampling methods to create a more balanced dataset for training our classifier model.

4.2 Resampling

We will use oversampling to increase the number of instances in the minority class (unreliable respondents) by generating synthetic samples. The technique for oversampling is SMOTE (Synthetic Minority Over-sampling Technique), which generates synthetic instances of the minority class by interpolating between existing instances.

```
set.seed(seed)
# Apply SMOTE to the training set
train_data_smote <- SMOTE(X = train_data[, -21], target = train_data[,
    21], dup_size = 8)
train_data_smote <- train_data_smote$data

# Check the class distribution after oversampling
table(train_data_smote$class)
```

```
##
```

```
##    0    1
## 6579 5481
```

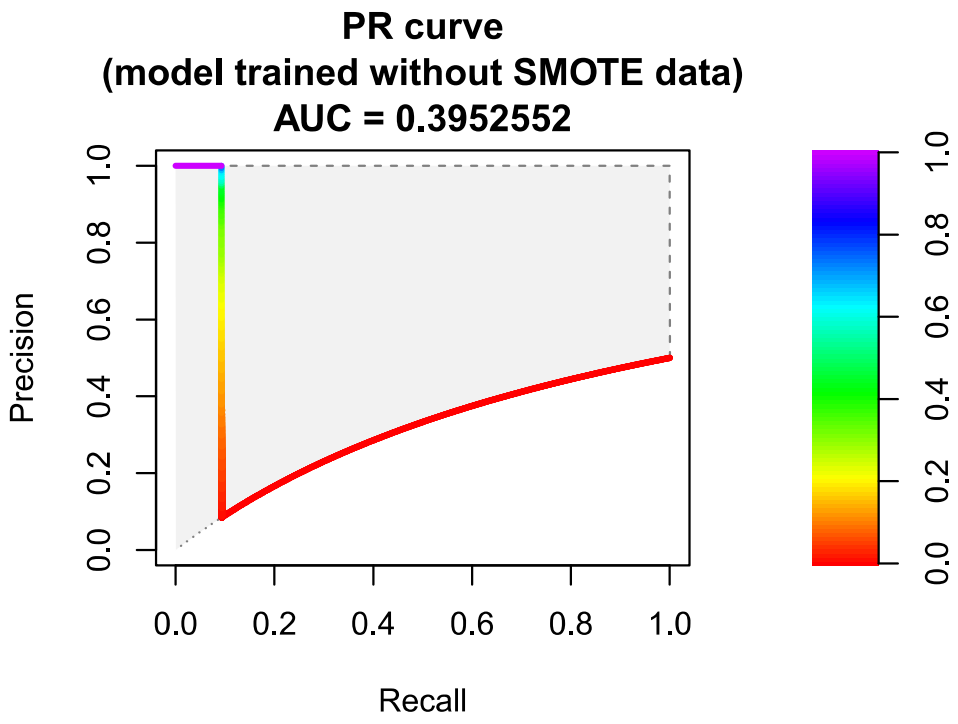
After applying SMOTE to the training set using the `smotefamily` library, we can see that the class distribution is now more balanced, with a similar number of instances in both the reliable and unreliable classes. Since the training set has changed, we need to retrain the model and evaluate its performance.

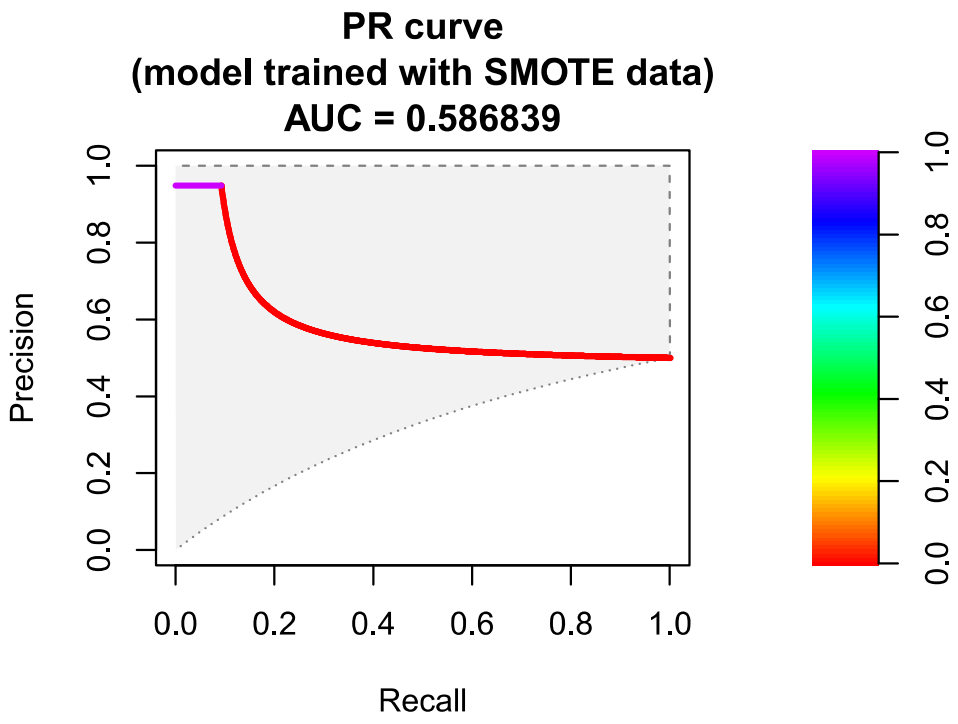
5 Results

Resampling methods such as SMOTE can enhance model performance metrics, including ROC and accuracy, particularly on balanced training data. However, there is a risk that the model may underperform when applied to imbalanced data. This can occur because the model is trained on data where the class proportions have been artificially adjusted, leading it to learn patterns that may not exist in the actual data distribution.

To address this issue, we plan to apply different evaluation metrics that are less sensitive to class imbalance, such as the area under the Precision-Recall curve. These metrics place more emphasis on the positive (minority) class, potentially making them a more suitable choice for our data analysis.

```r
# Create a PR curve object
pr2 <- pr.curve(test_data$class, test_pred_labels, curve = TRUE,
    max.compute = TRUE, min.compute = TRUE)

# Plot the PR curve
plot(pr, max.plot = TRUE, min.plot = TRUE, fill.area = TRUE,
    main = "PR curve\n(model trained without SMOTE data)")
plot(pr2, max.plot = TRUE, min.plot = TRUE, fill.area = TRUE,
    main = "PR curve\n(model trained with SMOTE data)")
```





After implementing the resampling technique using SMOTE to address the imbalance in the dataset, we observed an improvement in the classifier model's performance. The AUC-PR scores increased from 0.3952552 to 0.586839, indicating that the model is now better at identifying unreliable respondents.

6 Discussion

Strengths

1. Large Sample Size: The dataset contains a large number of samples, which provides us with sufficient data to train and evaluate our classifier model. The larger the sample size, the better the model can learn the underlying patterns and relationships between the variables, resulting in better predictive performance.

2. Rich Feature Set: The dataset includes a wide range of variables, capturing various aspects of the respondents' characteristics and behavior. This comprehensive feature set allows us to explore and identify the most relevant and informative features for predicting unreliable responses.

3. Established Data Collection Method: The survey data has been collected using a well-established and standardized data collection method, which ensures that the data is reliable and consistent across different samples and time periods.

Limitations

1. Imbalanced Data: The dataset has a imbalance between the reliable and unreliable classes. This imbalance can lead to biased predictions, as the classifier model may prioritize predicting the majority class (reliable respondents) over the minority class (unreliable respondents).

2. Potential Overfitting: Given the large number of features in the dataset, there is a risk of overfitting, which occurs when the model learns the noise in the training data rather than the underlying patterns.

3. Noisy Labels: It's possible that some respondents could be mislabeled as reliable or unreliable due to the inherent uncertainty in judging respondent reliability. Mislabeling in training data can lead to errors in model predictions.

4. Limited Ground Truth: The determination of unreliable respondents is often based on subjective judgment or certain predefined rules, which may not always accurately reflect the true reliability of a respondent. Therefore, the model's ability to generalize to new data could be affected if these rules do not apply universally.